

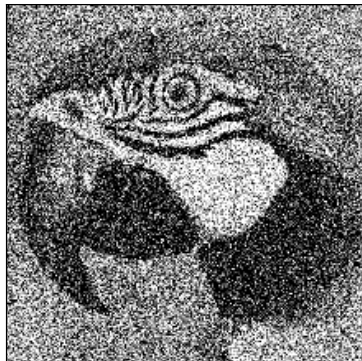
AI-powered Denoising: Supervised Learning

The denoising problem

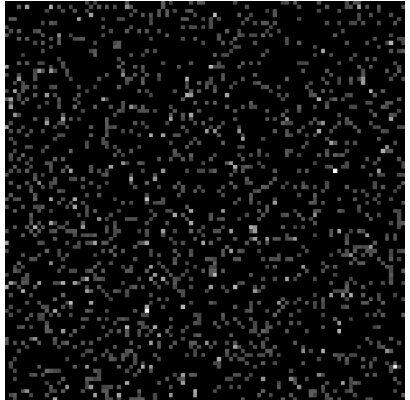
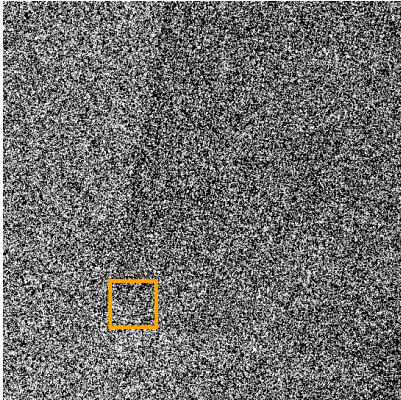
Estimate this



From this

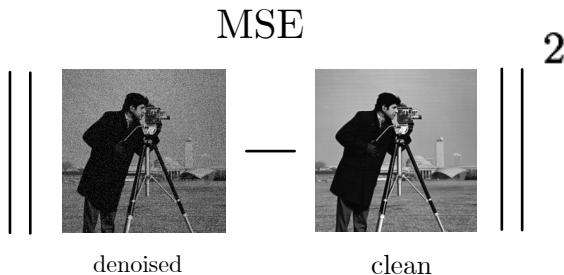


Why?



How do we evaluate a denoiser?

Mean squared error

$$\left\| \begin{array}{c} \text{denoised} \\ \text{clean} \end{array} \right\|_{\text{MSE}}^2$$


Peak signal-to-noise ratio

$$\text{PSNR} := 10 \log \left(\frac{M^2}{\text{MSE}} \right)$$

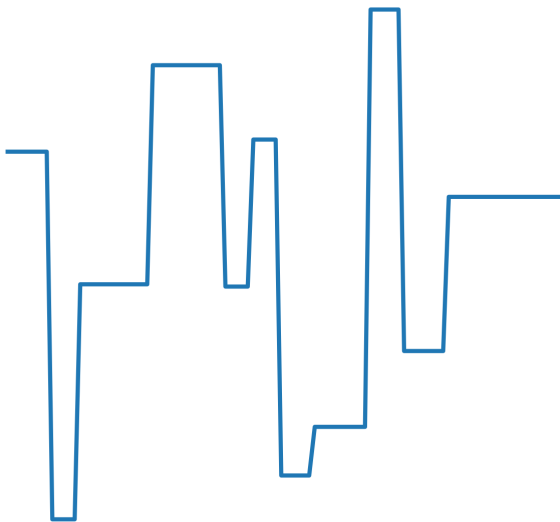
M : peak signal value

Warning: May not reflect **application-specific** performance

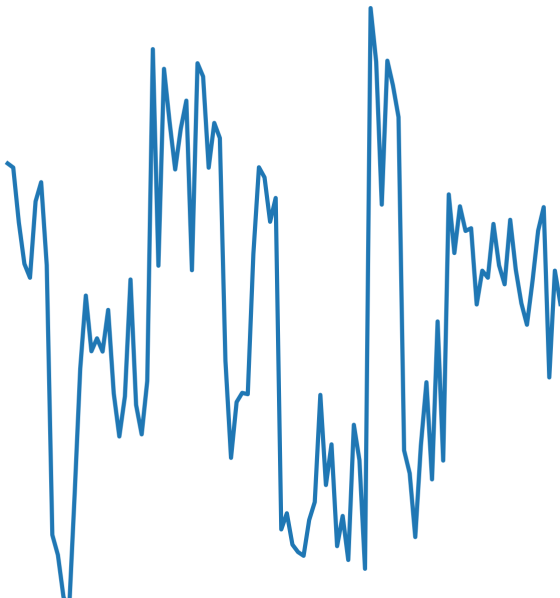
Example datasets

1. 1D piecewise-constant functions
2. Natural photographic images
3. Simulated electron-microscopy data

1D piecewise-constant functions



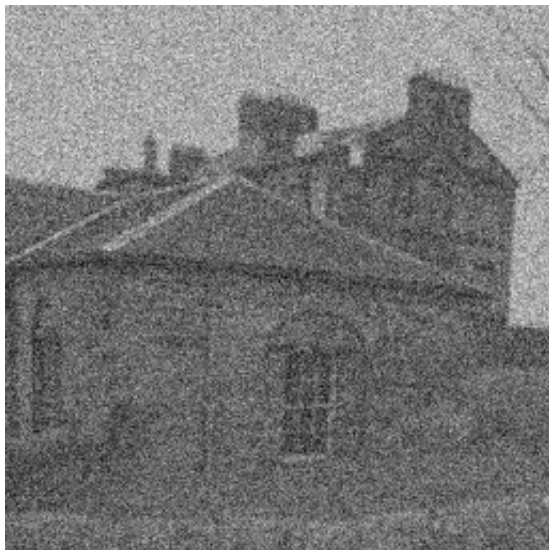
Additive i.i.d. Gaussian noise (PSNR 19.96 dB)



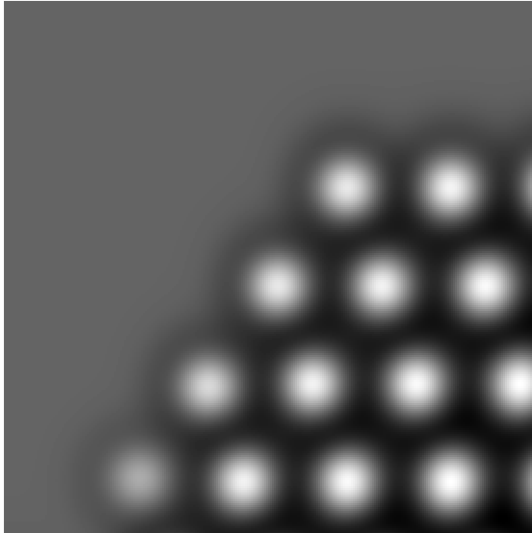
Natural images



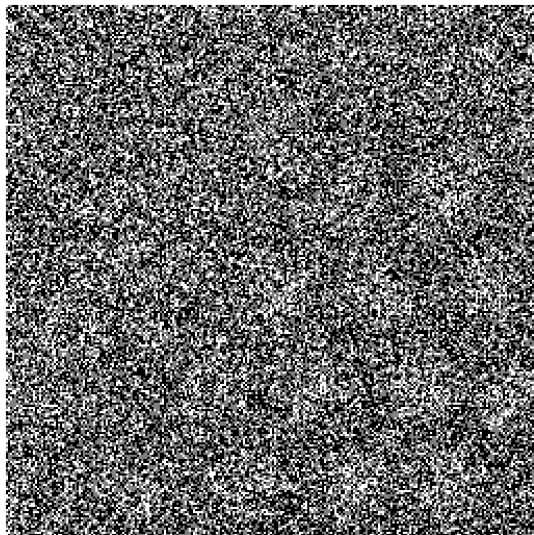
Additive i.i.d. Gaussian noise with variable variance (PSNR 19.99 dB)



Electron microscopy



Poisson shot noise (PSNR 6.64 dB)



Plan

Denoising before deep learning

Convolutional neural networks

Supervised denoising

What do deep denoisers learn?

Conclusions (and a word of caution!)

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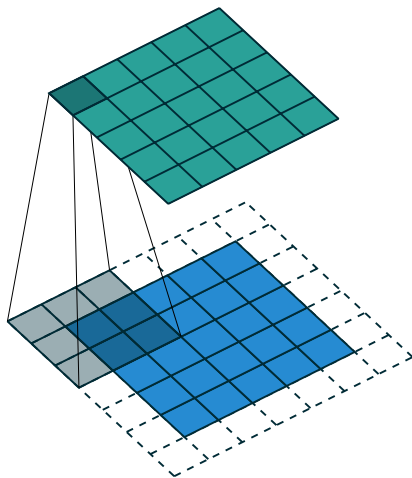
Challenge

How to estimate thousands of clean pixels from thousands of noisy pixels?

Idea: Assume (1) linearity and (2) shift invariance

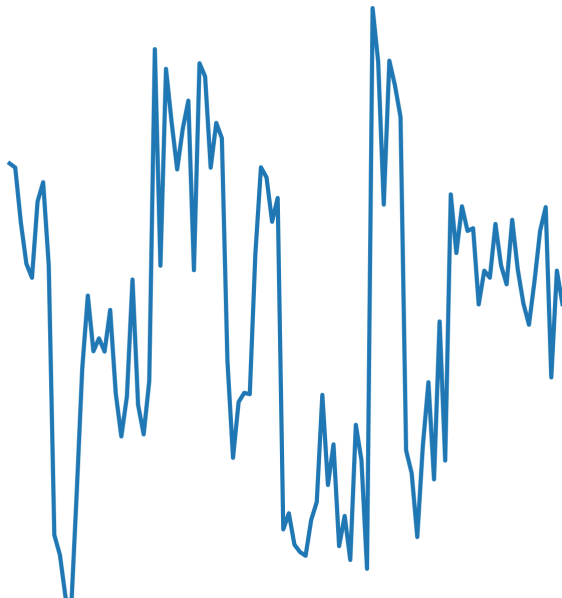
Optimal (Wiener) denoiser is convolutional

Convolutional filter

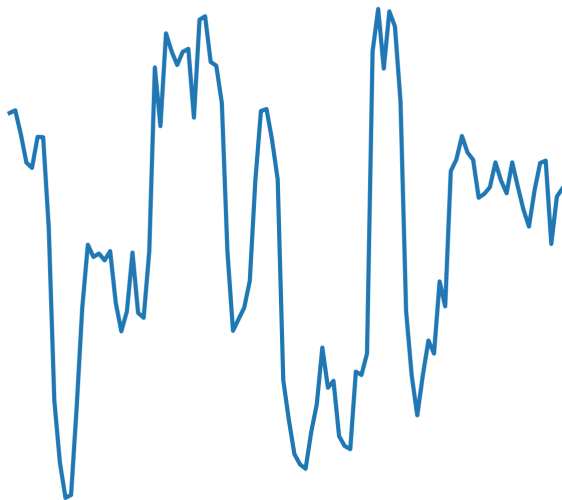


A guide to convolution arithmetic for deep learning, Dumoulin & Visin, 2016

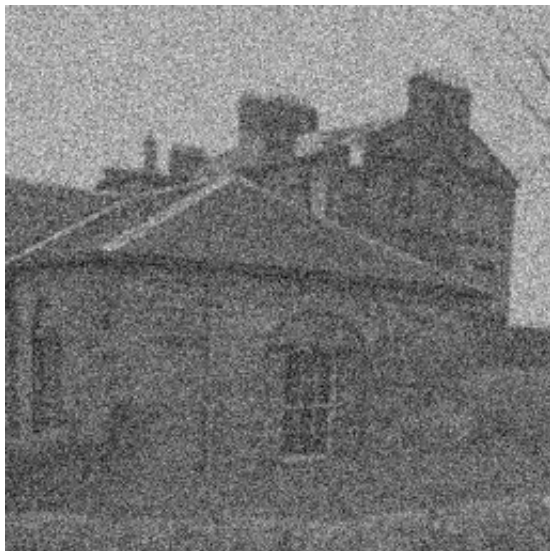
Noisy data (PSNR 19.96 dB)



LSI denoising (PSNR 22.33 dB)



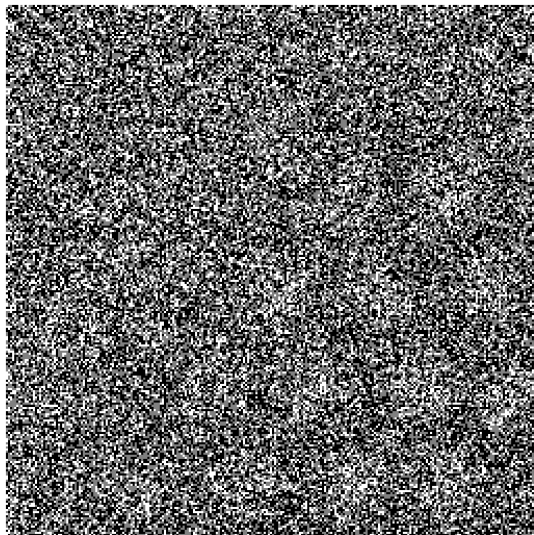
Noisy data (PSNR 19.99 dB)



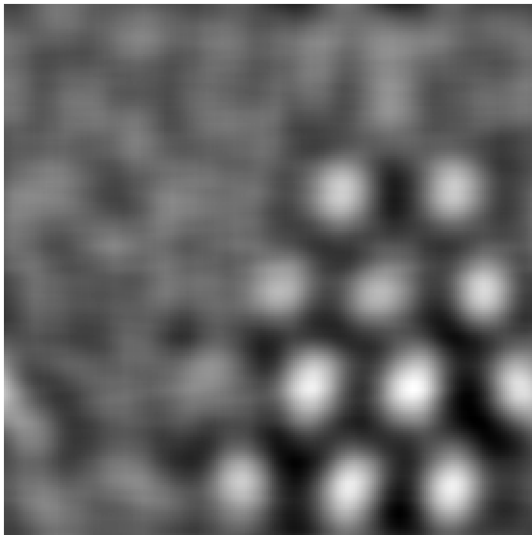
LSI denoising (PSNR 25.78 dB)



Noisy data (PSNR 6.64 dB)



LSI denoising (PSNR 29.80 dB)



Limitations of LSI denoising?

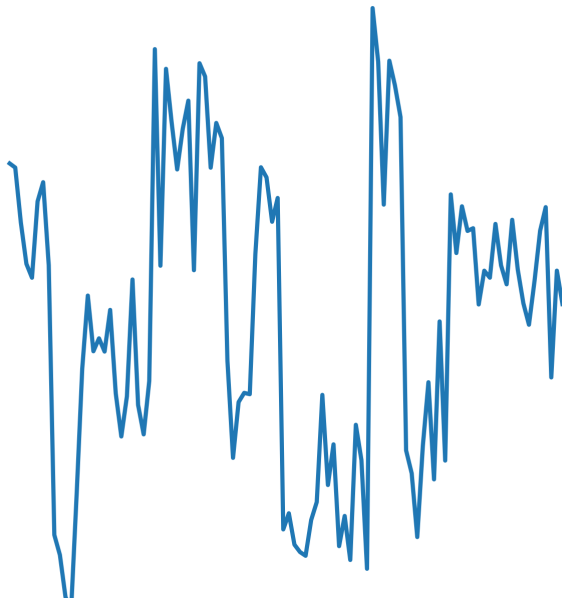
Same filter applied at every location

Blurs edges and other features

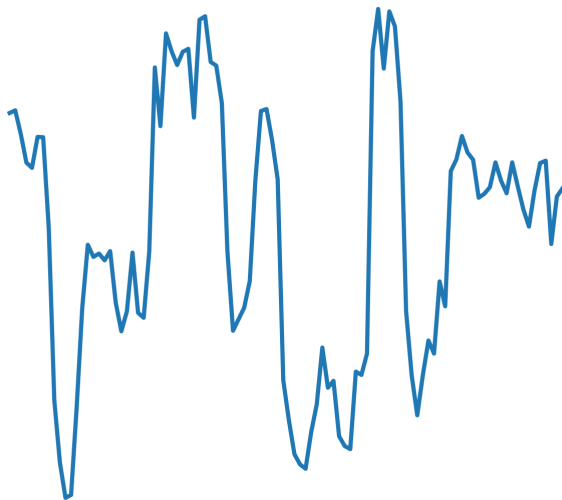
Pre-deep-learning solutions

- ▶ Adapt filter locally (e.g. bilateral filter)
- ▶ Exploit approximate sparsity of images in transformed domains (finite differences, wavelets...)
- ▶ Exploit similarity between image patches to perform nonlocal averaging or collaborative filtering

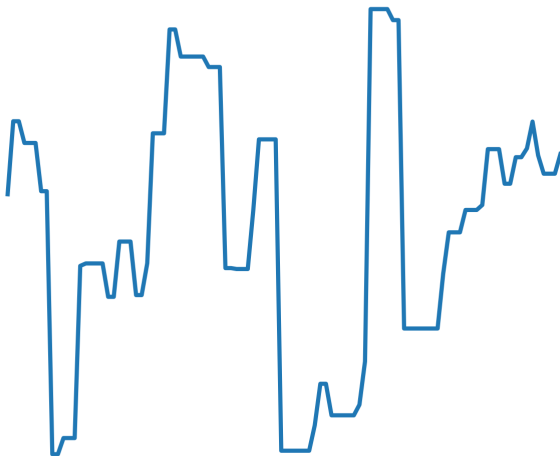
Noisy data (PSNR 19.96 dB)



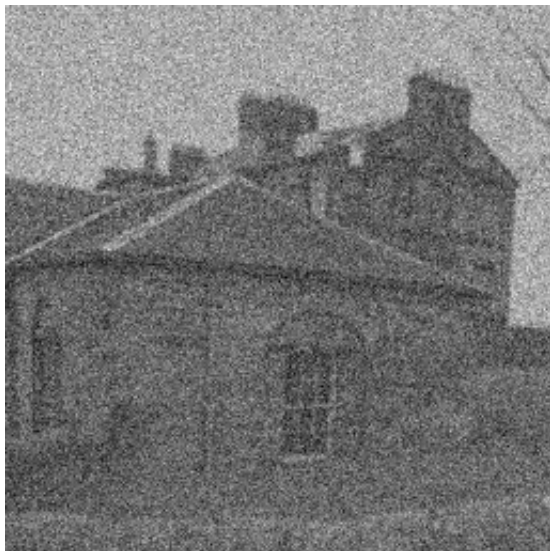
LSI denoising (PSNR 22.33 dB)



Total variation denoising (PSNR 25.07 dB)



Noisy data (PSNR 19.99)



LSI denoising (PSNR 25.78)

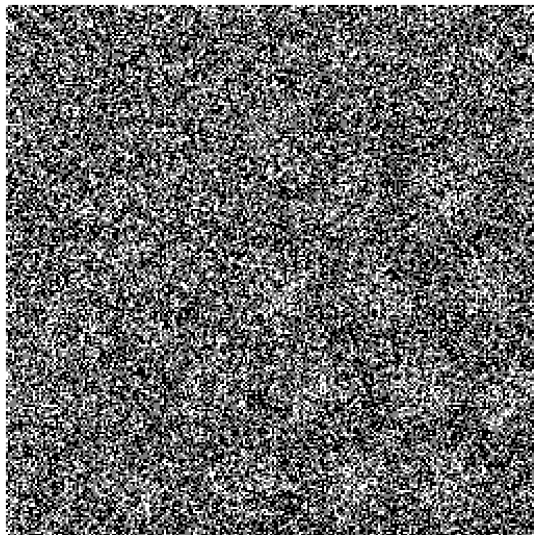


BM3D denoising (PSNR 30.62)

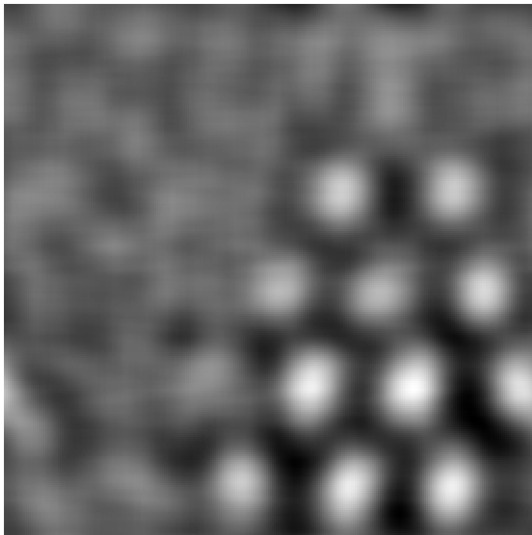


Image denoising by sparse 3-D transform-domain collaborative filtering. Dabov et al. 2007

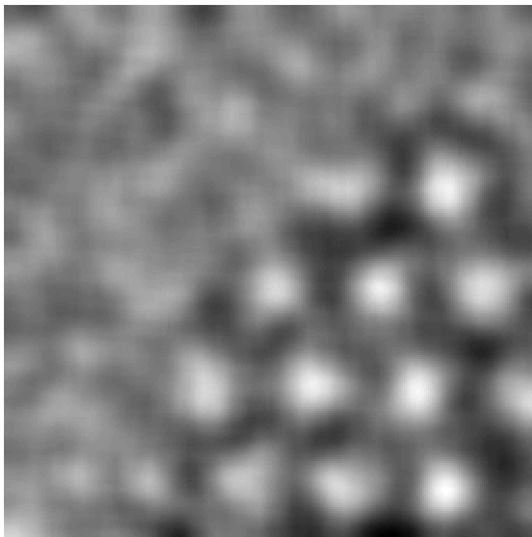
Noisy data (PSNR 6.64)



LSI denoising (PSNR 29.80)



PURE-LET denoising (PSNR 30.72)



A new SURE approach to image denoising: Interscale orthonormal wavelet thresholding.
Luisier et al. 2007

Denoising before deep learning

Convolutional neural networks

Supervised denoising

What do deep denoisers learn?

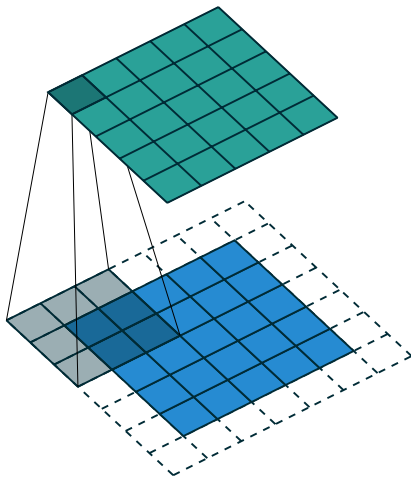
Conclusions (and a word of caution!)

Convolutional neural networks

Designed to learn complex nonlinear functions from data

Convolutional layers interleaved with nonlinearities

Convolution



A guide to convolution arithmetic for deep learning, Dumoulin & Visin, 2016

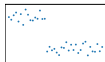
ReLU

Rectifying linear unit

Sets **negative** entries to **zero**

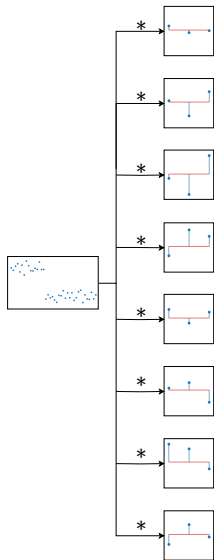
Convolutional neural network

Input



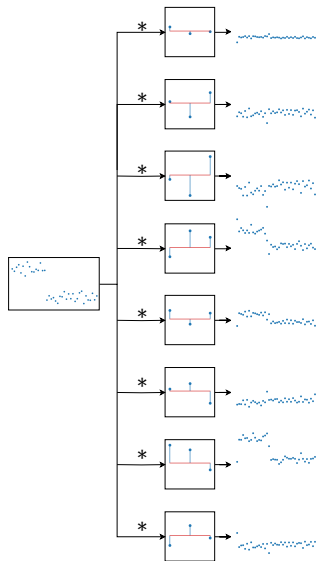
Convolutional neural network

Input Convolutional layer



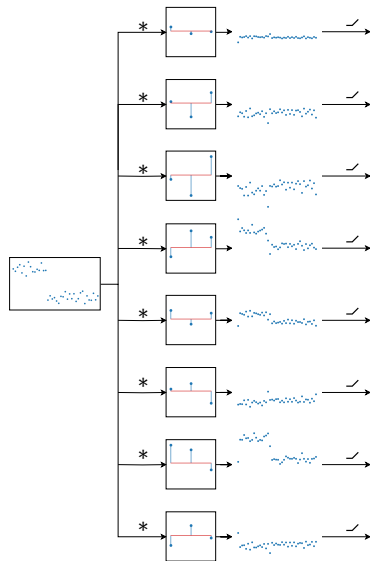
Convolutional neural network

Input Convolutional layer



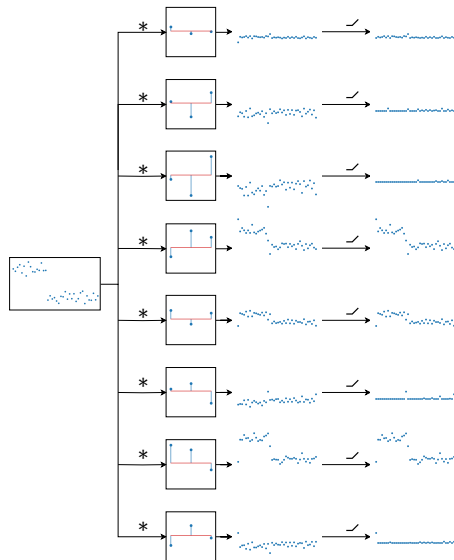
Convolutional neural network

Input Convolutional layer ReLU



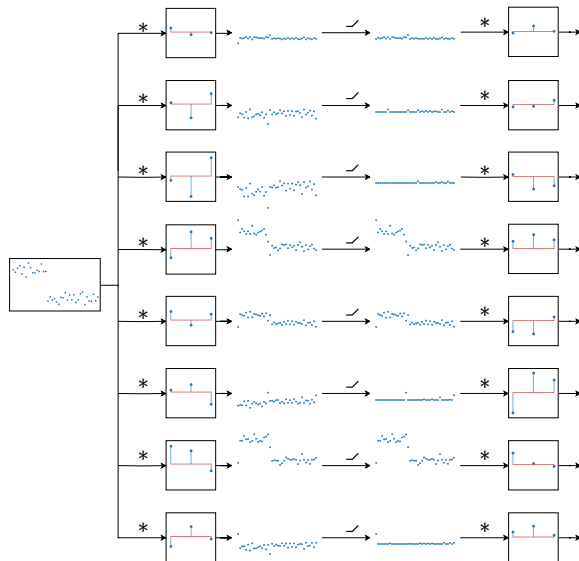
Convolutional neural network

Input Convolutional layer ReLU



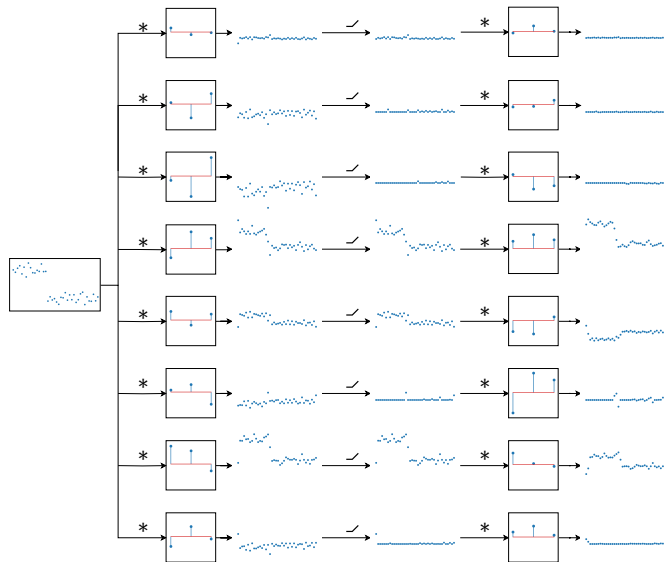
Convolutional neural network

Input Convolutional layer ReLU Convolutional layer



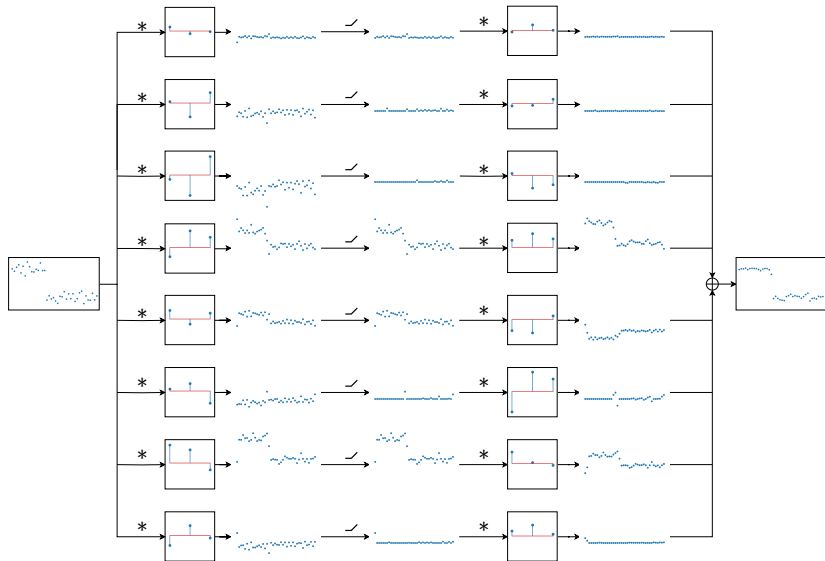
Convolutional neural network

Input Convolutional layer ReLU Convolutional layer



Convolutional neural network

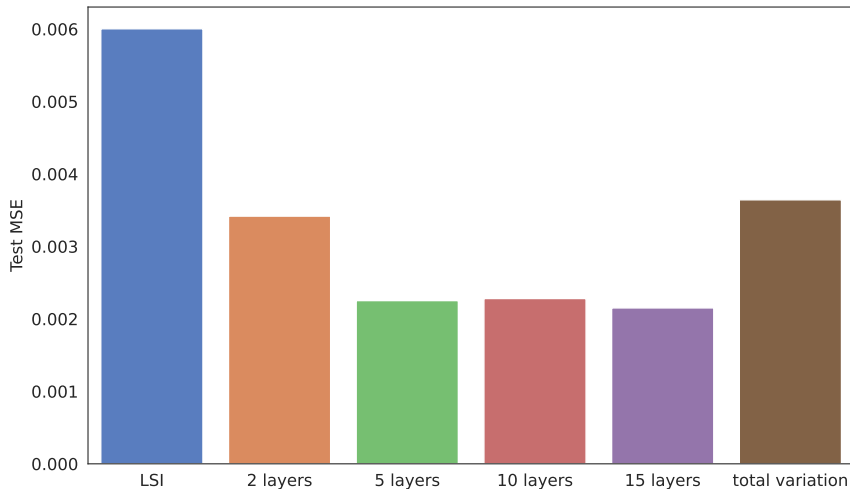
Input Convolutional layer ReLU Convolutional layer Output



Number of layers?

Too few layers can result in underfitting

Once the network is complex enough, diminishing returns



Batch normalization

Features in each channel are

1. Standardized, removing mean and dividing by standard deviation
2. Re-centered and re-scaled with learned parameters

Skip connections

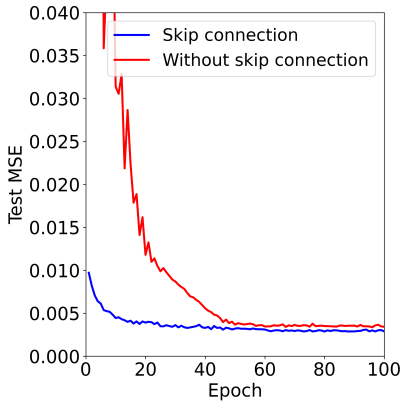
Links between layers at different depths, bypassing the layers in between

In denoising, a skip connection between input and output is often helpful

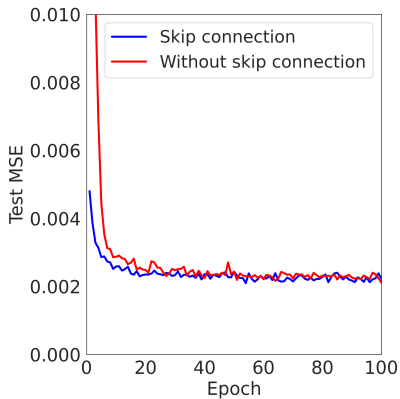
Equivalently, we learn a **residual** that is added to the noisy input

Piecewise constant signals

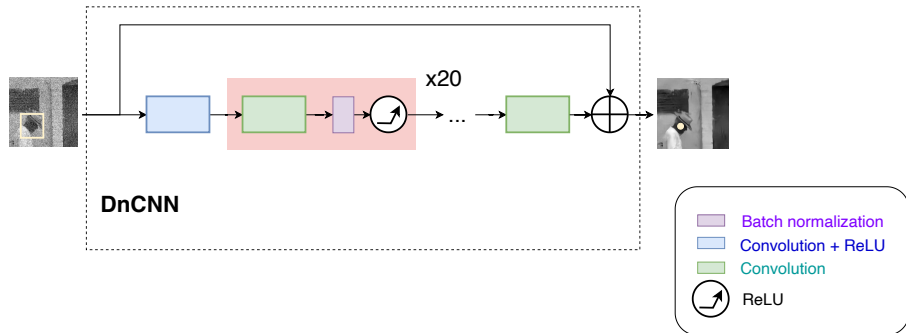
Small training set



Large training set

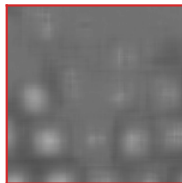
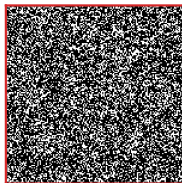
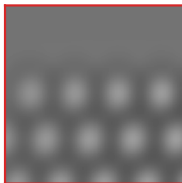
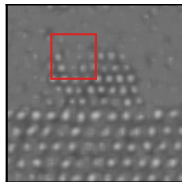
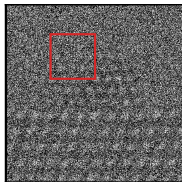
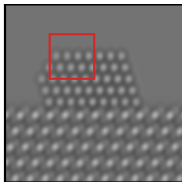


Denoising CNN (DnCNN)



Beyond a Gaussian Denoiser: Residual Learning of Deep CNN for Image Denoising.
Zhang et al. 2017

Electron microscopy

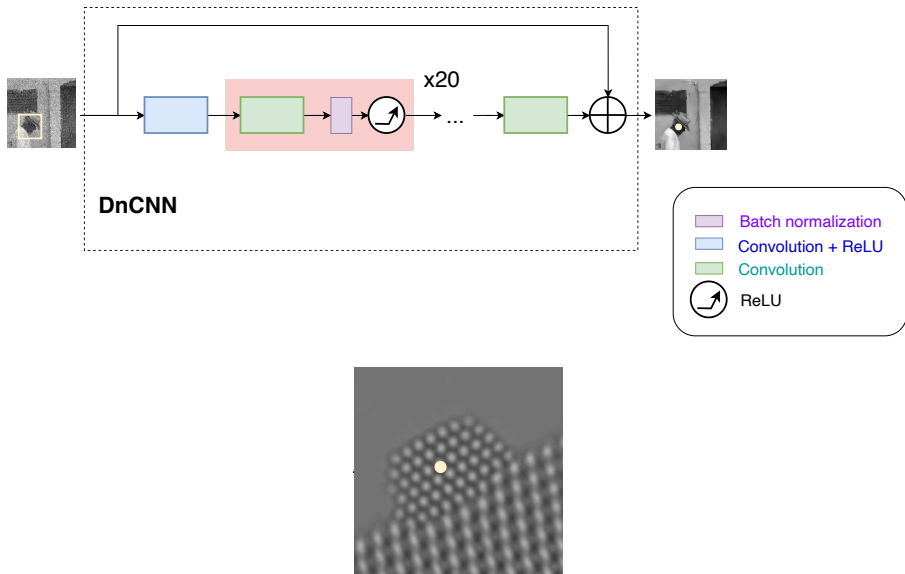


Simulated
clean image

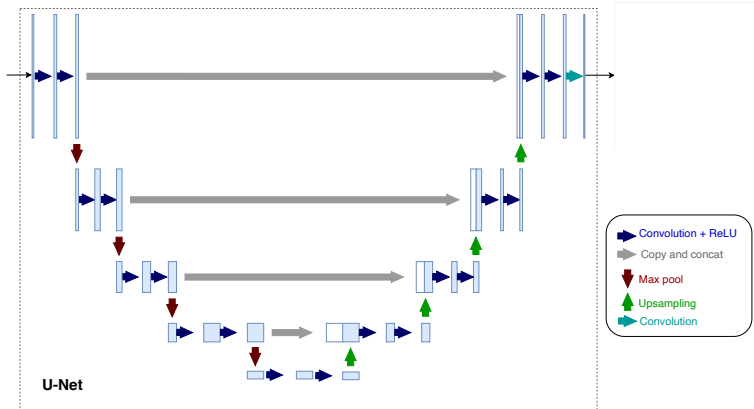
Noisy image

DnCNN

Receptive field

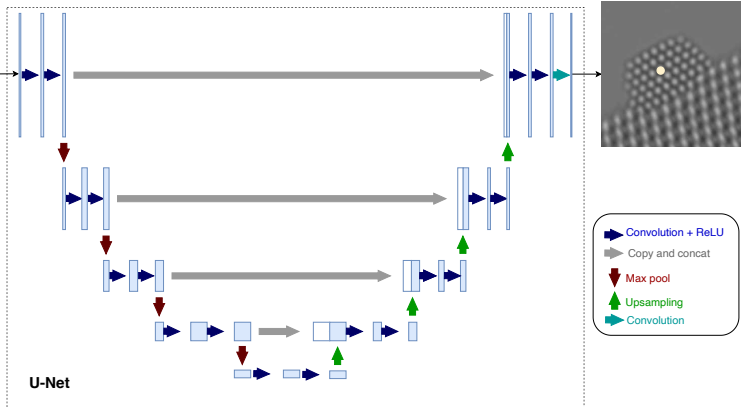
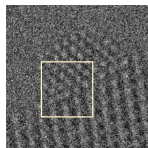


U-Net

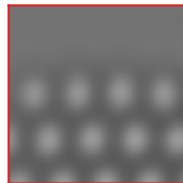
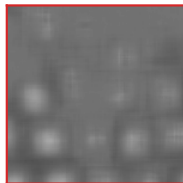
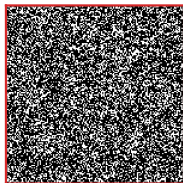
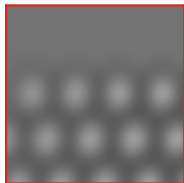
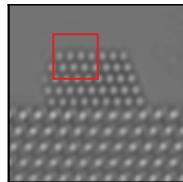
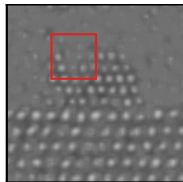
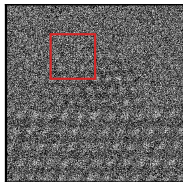
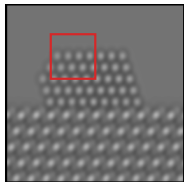


U-Net: convolutional networks for biomedical image segmentation. Ronneberger et al. 2015

U-Net



Electron microscopy



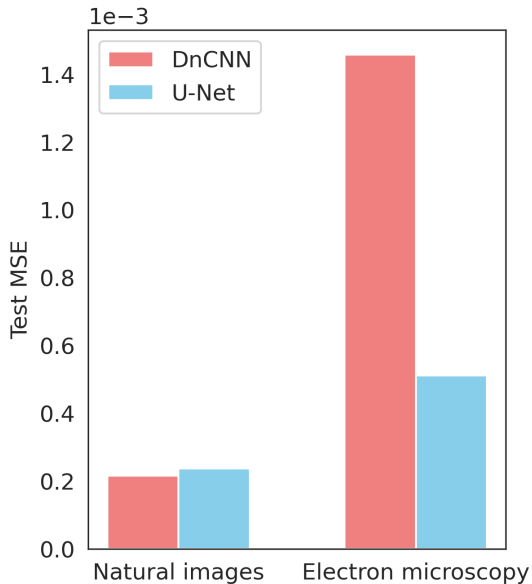
Simulated
clean image

Noisy image

DnCNN

UNet

DnCNN vs U-Net



Denoising before deep learning

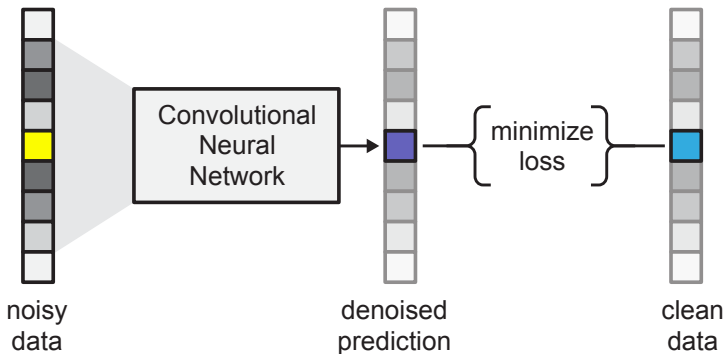
Convolutional neural networks

Supervised denoising

What do deep denoisers learn?

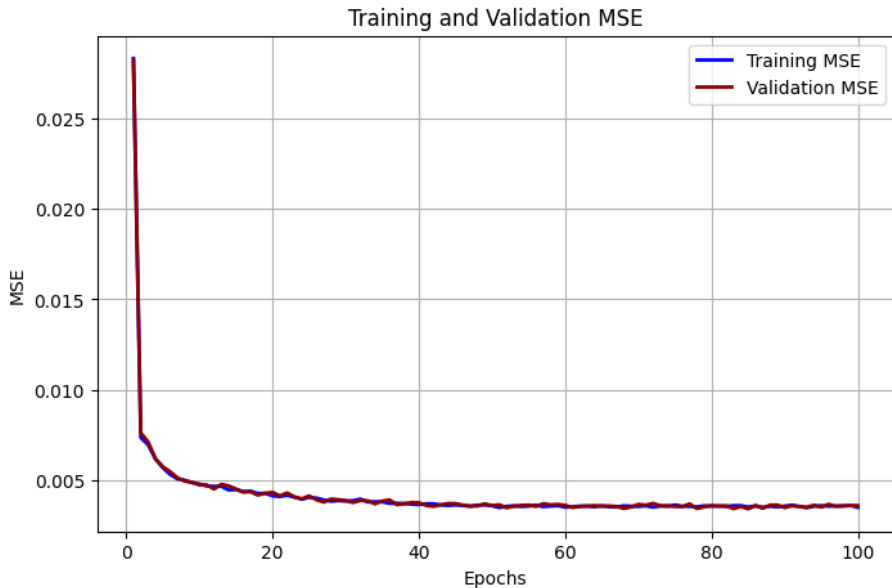
Conclusions (and a word of caution!)

Supervised learning



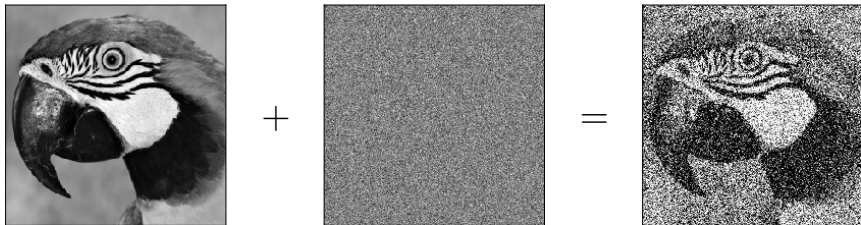
- ▶ Assumes access to clean - noisy pairs
- ▶ Minimizes mean squared error over training pairs via stochastic gradient descent
- ▶ Gradient updates computed via backpropagation

Piecewise constant functions



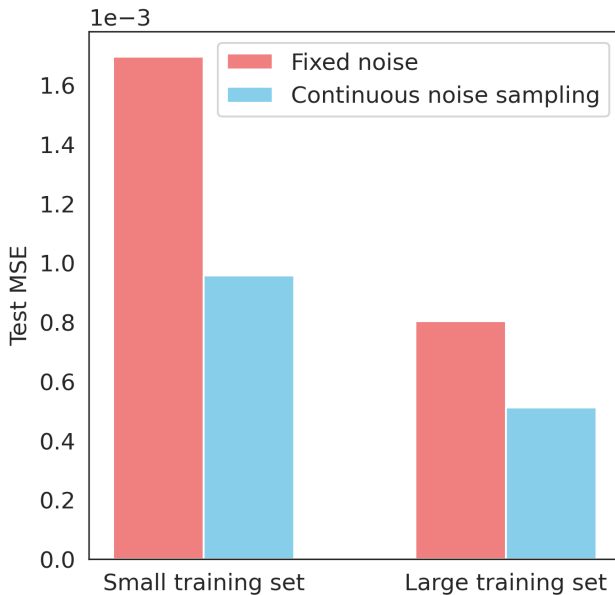
Continuous noise sampling

If we know the noise distribution, we can simulate it **continuously** during training



Prevents **overfitting** specific noise samples

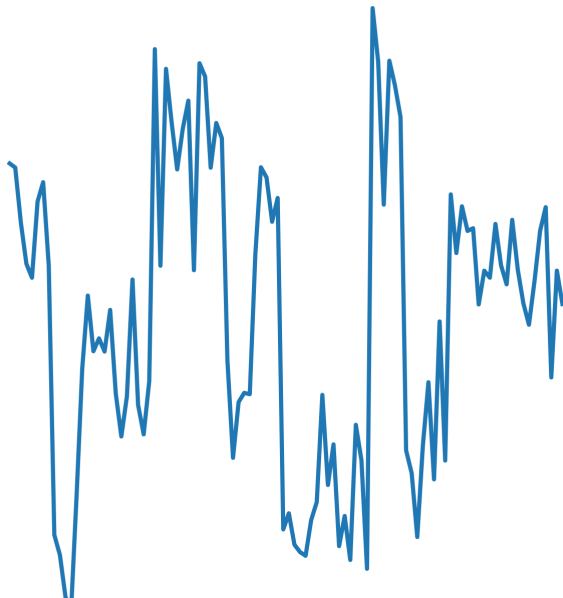
Electron microscopy data



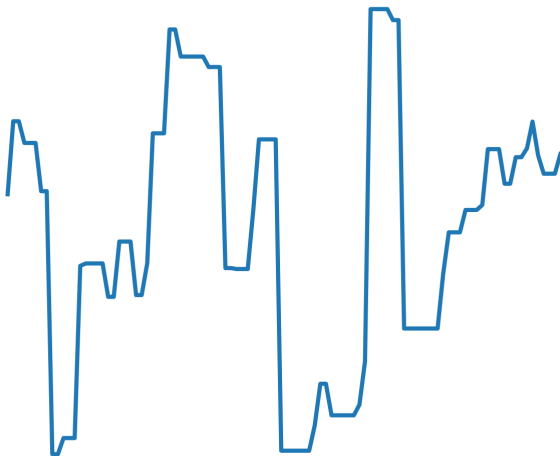
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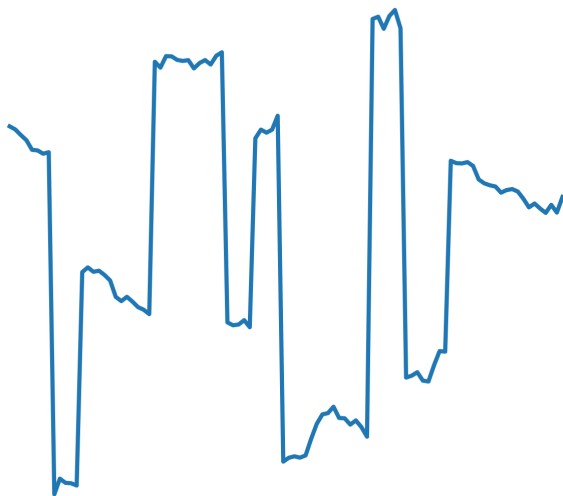
Noisy data (PSNR 19.96 dB)



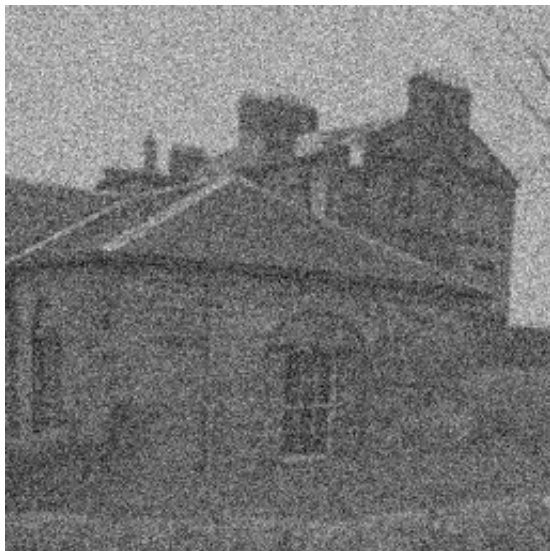
Total variation denoising (PSNR 25.07 dB)



Neural network (PSNR 26.44 dB)



Noisy data (PSNR 19.99 dB)



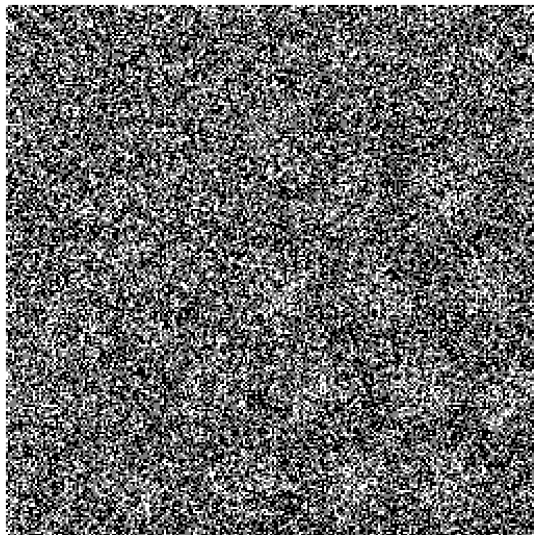
BM3D denoising (PSNR 30.62 dB)



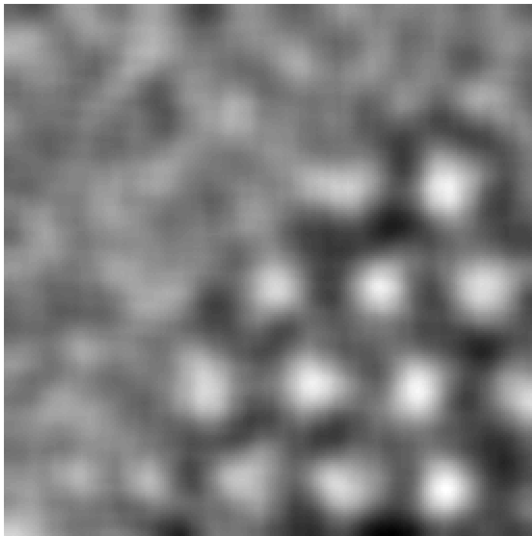
Neural network (PSNR 30.78 dB)



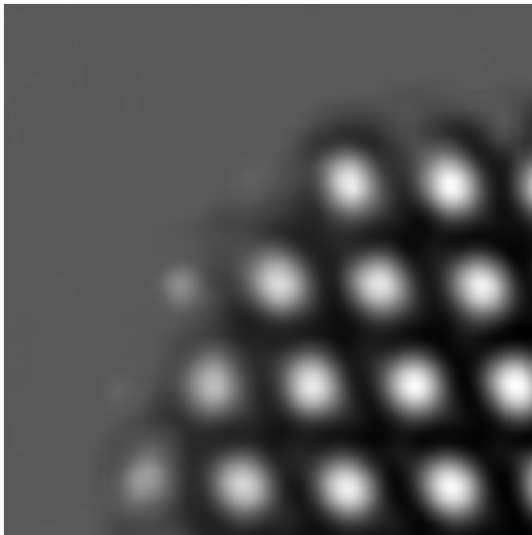
Noisy data (PSNR 6.64 dB)



PURE-LET denoising (PSNR 30.72 dB)



Neural network (PSNR 34.78 dB)



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Conclusions (and a word of caution!)

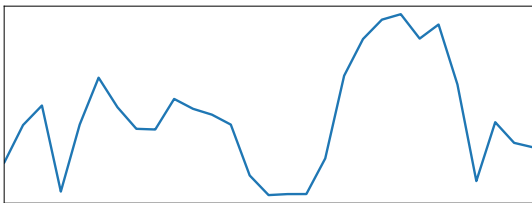
How do neural networks denoise?

They implement complex nonlinear functions that depend on the input

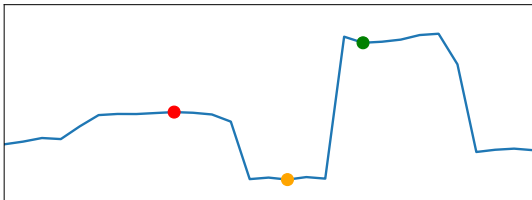
Gradient of output with respect to input reveals learned denoising mechanism *for that specific input*

Robust and interpretable blind image denoising via bias-free convolutional neural networks. Mohan et al. 2020

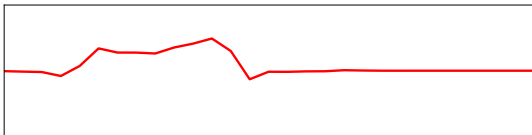
Noisy data



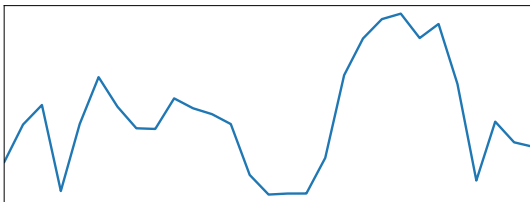
Denoised estimate



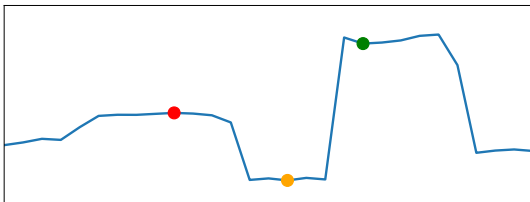
Gradient with respect to input



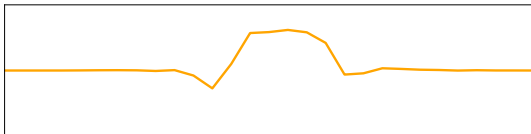
Noisy data y



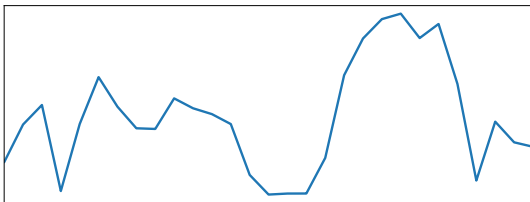
Denoised estimate $\mathcal{D}_{\Theta}(y)$



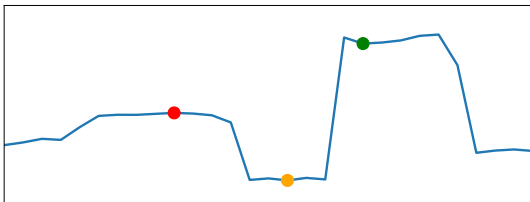
Gradient with respect to input



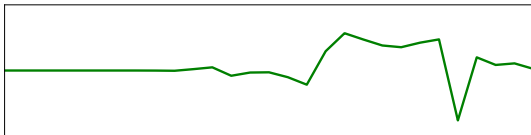
Noisy data y



Denoised estimate $\mathcal{D}_{\Theta}(y)$

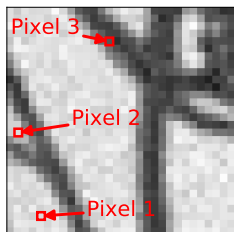


Gradient with respect to input

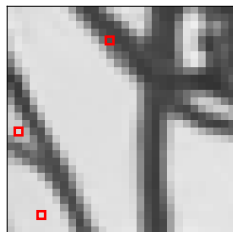


Low noise

Noisy image



Denoised



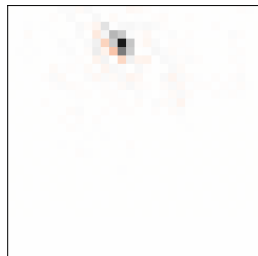
Pixel 1



Pixel 2

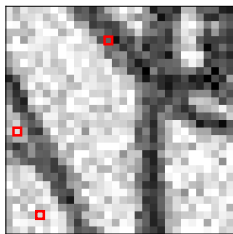


Pixel 3

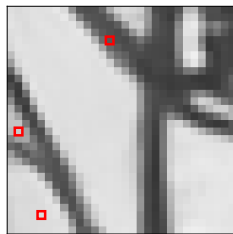


Medium noise

Noisy image



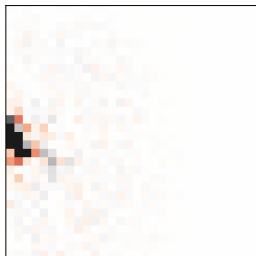
Denoised



Pixel 1



Pixel 2

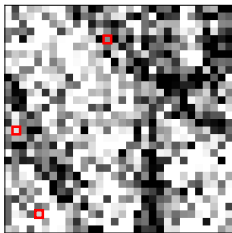


Pixel 3

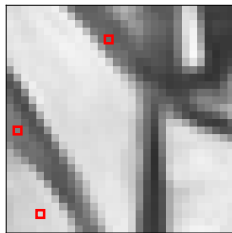


High noise

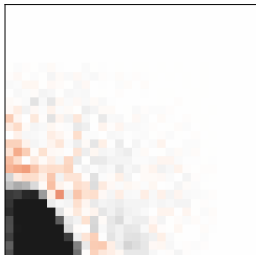
Noisy image



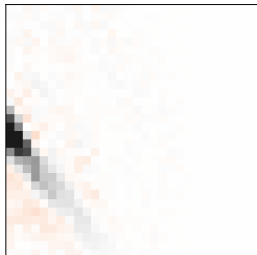
Denoised



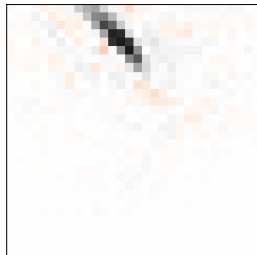
Pixel 1



Pixel 2

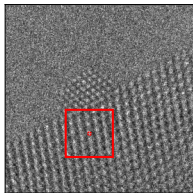


Pixel 3

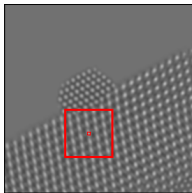


Electron microscopy

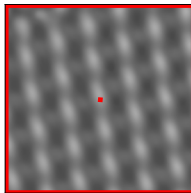
Noisy



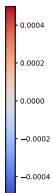
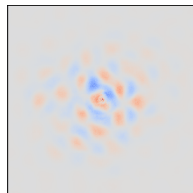
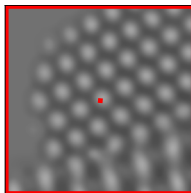
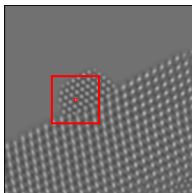
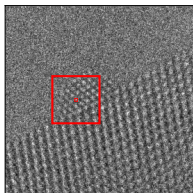
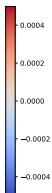
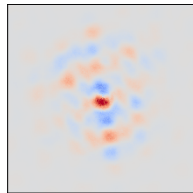
Denoised



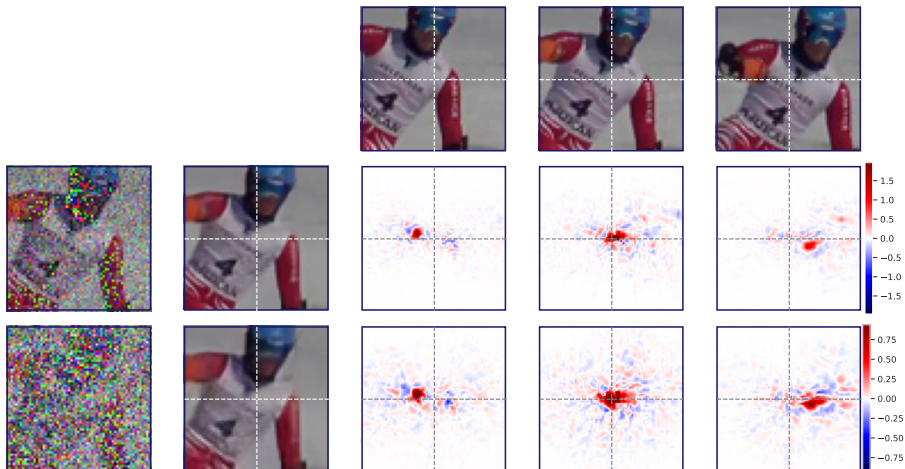
Denoised
(zoomed)



Gradient



Video denoising



Implicit motion compensation

(a) Noisy frame ($\sigma = 30$)



(b) Motion estimate from clean video



(c) Motion estimate from network gradients



Denoising before deep learning

Convolutional neural networks

Supervised denoising

What do deep denoisers learn?

Conclusions (and a word of caution!)

Conclusions

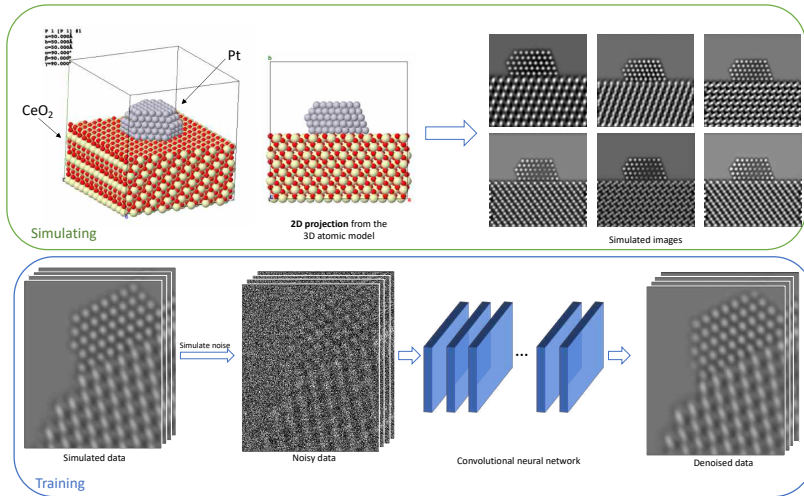
Supervised denoising is *extremely effective*

Models learn denoising strategies adapted to different signals and noise distributions **automatically**

A word of caution

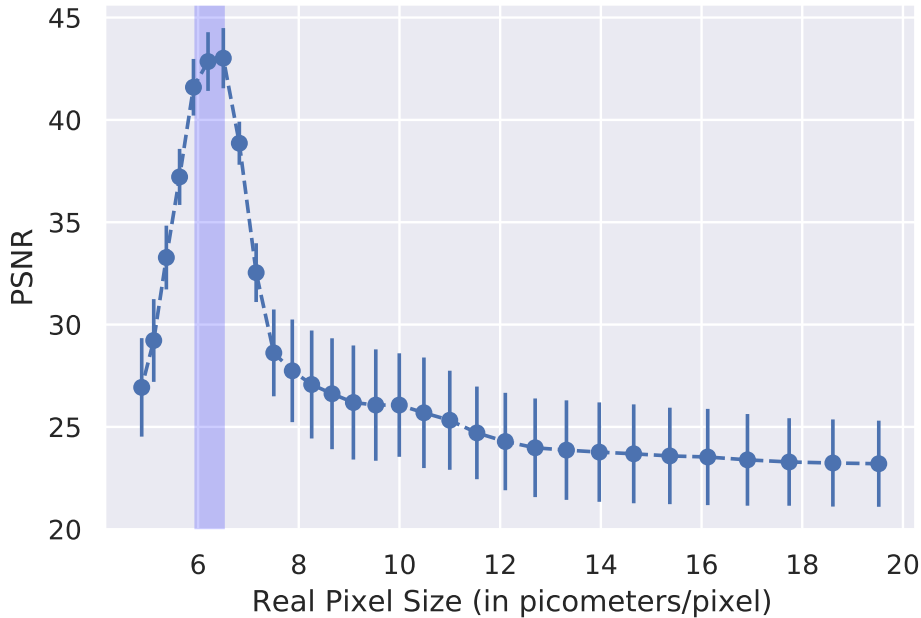
Distribution shift with respect to training data can cause models to fail!

Electron microscopy

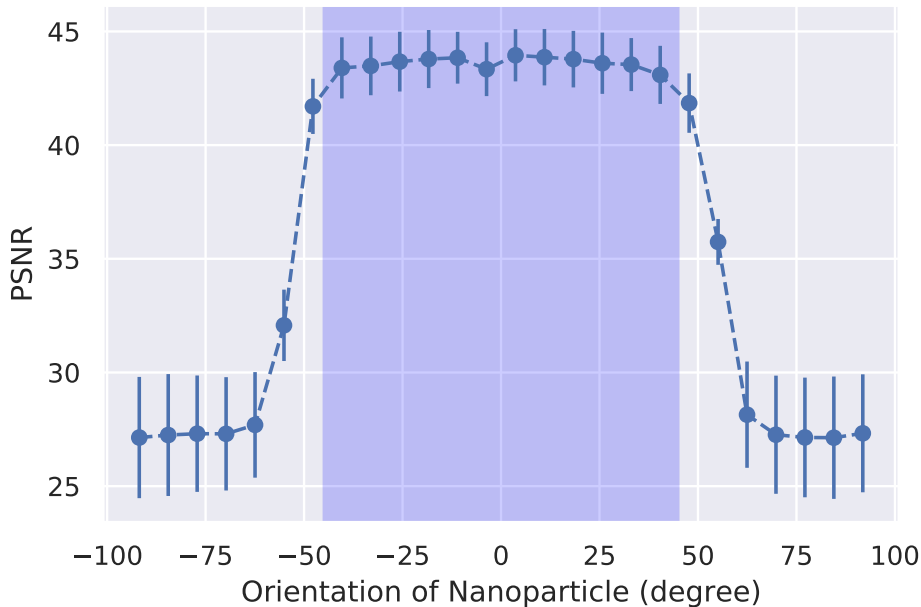


Experiment: Apply model to data at a different (1) resolution or (2) orientation

Resolution



Orientation



For more information

Learn, Denoise and Discover: A Guide to Deep Denoising with an Application to Electron Microscopy

Sreyas Mohan, Kangning Liu, Peter Crozier,
Carlos Fernandez-Granda

Acknowledgements

We are very grateful for the support of the [National Science Foundation](#) through grants OAC-2104105 and OAC-2103936